

Battery Ageing Model: Combined Calendar & Cycling Implementation

To Project Supervisor
Re Battery State-of-Health model, 2-year simulation
Reference Nájera et al., *J. Energy Storage* 72 (2023) 108016

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Objective

Replace the previous time-only degradation model with a combined **calendar + cycling ageing model**, so battery health predictions depend on current (charge/discharge), State of Charge, and temperature — not elapsed time alone.

The two mechanisms are modelled separately and then summed, matching the source paper's own construction: a battery loses capacity both from *sitting* (calendar ageing) and from *being used* (cycling ageing).

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Formulas implemented

● CALENDAR AGEING

$$\text{Ln}(Q_{\text{cal}}) = f \cdot e^{(g \cdot \text{SoC})} \cdot e^{(h/T)} \cdot t_{\text{days}}^z$$

Depends on **SoC**, constant **T**, elapsed **time**

● CYCLING AGEING

$$\text{Ln}(Q_{\text{cyc}}) = |a \cdot T^2 + b \cdot T + c| \cdot e^{((d \cdot T + e) \cdot \text{Crate})} \cdot \text{Ah}$$

Depends on **current** (via C-rate, Ah), constant **T**

Combined into a single State of Health: $\text{SOH} = 1 - (Q_{\text{cal}} + Q_{\text{cyc}}) / 100$

PARAMETER	VALUE	USED IN
f	647,260,000	Calendar

PARAMETER	VALUE	USED IN
g	1.4219	Calendar
h	−8,219.1	Calendar
z	0.5	Calendar
a	2.9961×10^{-8}	Cycling
b	-1.7339×10^{-5}	Cycling
c	0.0025	Cycling
d	−0.0124	Cycling
e	3.8738	Cycling
T	298.15 K (25 °C)	Both — held constant

Parameters correspond to the Sony US26650FT LFP cell — Table 3 of the reference paper — the only LFP cell there with both ageing mechanisms validated together.

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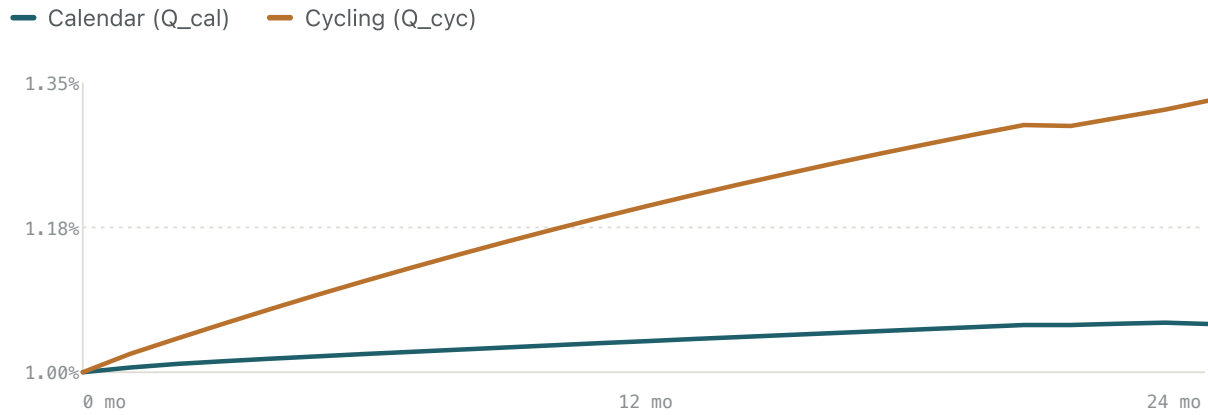
Files changed

main/Devices/Batteries.py	New function <code>Battery_SOH_Najera</code> implementing the combined model
main/BuilderBatteryPV.py	Registered the new model for use by the optimization builder
main/Cases/ExampleBatteryPV8760H_2Y.xlsx	2-year input dataset (1-year real data repeated twice)
main/ExampleBatteryPV8760.py	Runs the 2-year dataset through the new model; exports results
main/Cycling_and_Calendar_Aging/Battery_2Years_CombinedAging.csv	Output: hourly Power, SoC, Q_cal, Q_cyc, SOH over 2 years

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Results — 2-year simulation, 17,518 hourly steps

CALENDAR AGEING 1.000% → 1.019%	CYCLING AGEING 1.000% → 1.323%	STATE OF HEALTH 98.00% → 97.66%
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Over 24 months, cycling ageing accounts for the large majority of total capacity loss. This is consistent with the battery being held at low State of Charge for most of the horizon in this scenario — which the model correctly predicts as suppressing calendar ageing.

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Known limitation

NUMERICAL SENSITIVITY IN THE CYCLING TERM

The coefficient $(a \cdot T^2 + b \cdot T + c)$ is a near-cancellation of three very small numbers. Using only the ~5 significant figures published in the paper's table causes this term to flip sign at typical operating temperatures — an artifact not present in the paper's own figures. A magnitude-preserving correction (`abs()`) was applied and documented in code. Higher-precision parameters from the underlying PhD thesis (ref. [21] in the paper) would remove the need for this safeguard.